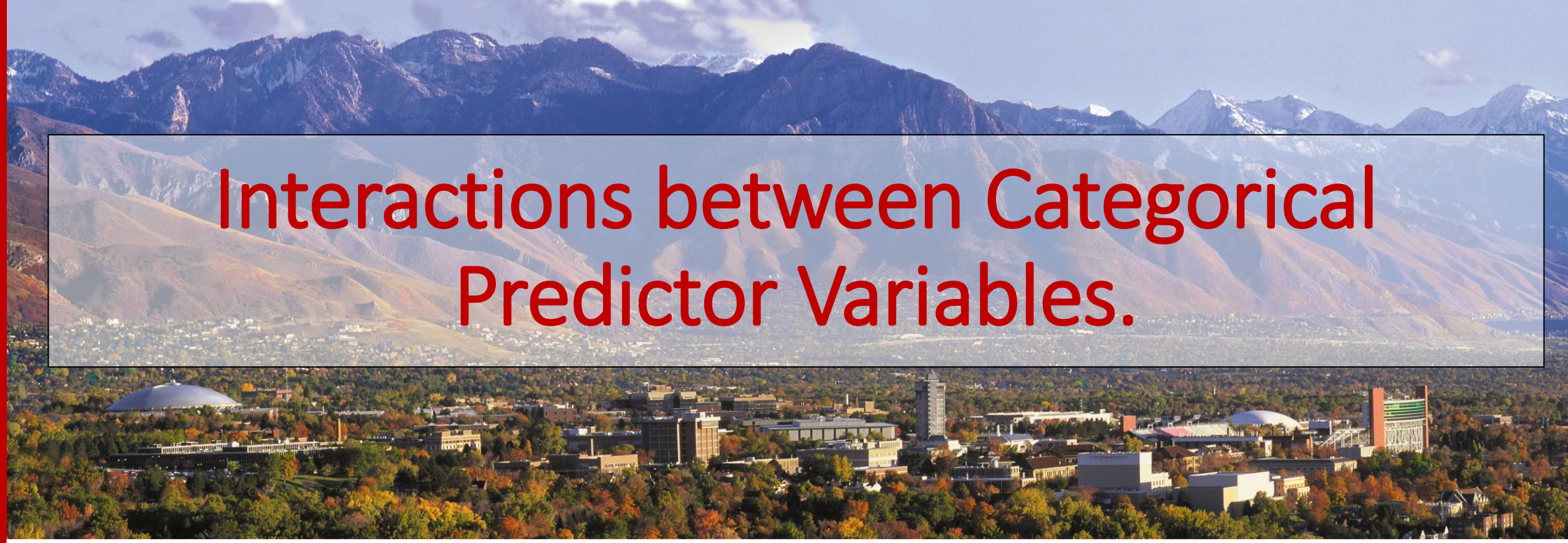

Interactions between Categorical Predictor Variables.

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HEALTH
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Two factor design example:
Placebo effects and gender.

- We induce controlled thermal pain to males and females (factor 1) who have taken a either placebo or nothing (factor 2).
- Each factor has two levels: 2 x 2 design

	Men	Women
Placebo	There are many people in each cell	Here, 30 females who took placebo
Nothing	Here, 30 males who took nothing	We'll put the *mean* score in each box

“Main effect” of gender

	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6
	6.4	6.5

Men vs. Women

“Main effect” of placebo

	Men	Women	
Placebo	5.3	6.4	5.85
Nothing	7.5	6.6	7.05
	6.4	6.5	

Placebo vs nothing

“Interaction” effect:

Does the effect of placebo *depend* on gender?

	Men	Women	
Placebo	5.3	6.4	5.85
Nothing	7.5	6.6	7.05
	6.4	6.5	

Difference of 2.2

Difference of 0.2

“Interaction” effect:

Does the effect of placebo *depend* on gender?

Are the differences different?
More correctly, does the effect of placebo change as a function of gender?

	Men	Women	
Placebo	5.3	6.4	5.85
Nothing	7.5	6.6	7.05
	6.4	6.5	

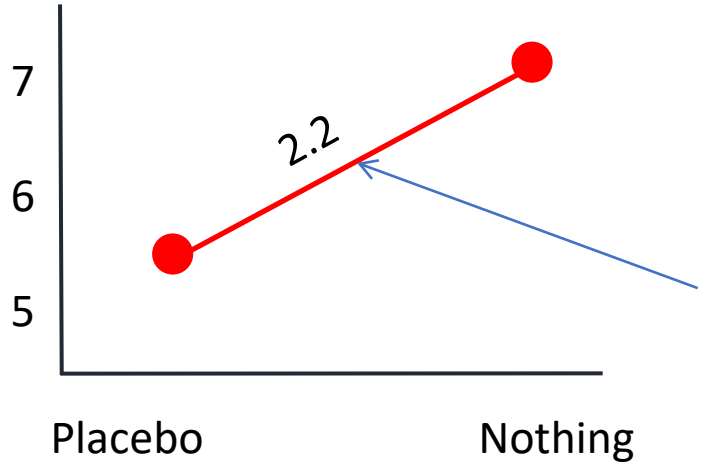
Difference of 2.2

Difference of 0.2

The difference between differences = 2.0

Plotting two-way factorial results

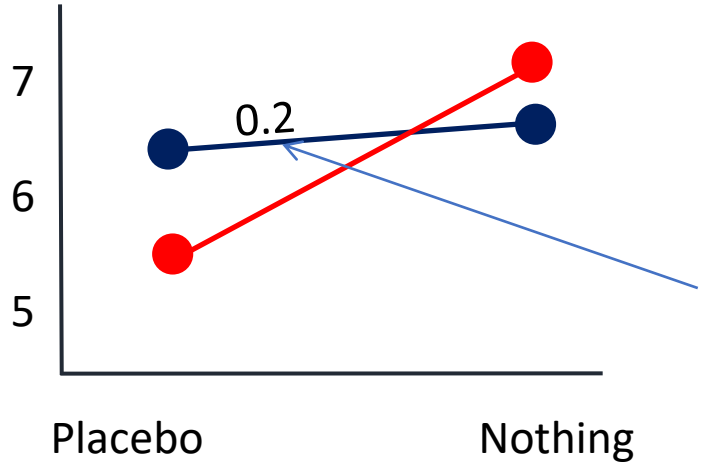
	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6



The slope of the red line is the placebo effect among men

Plotting two-way factorial results

	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6



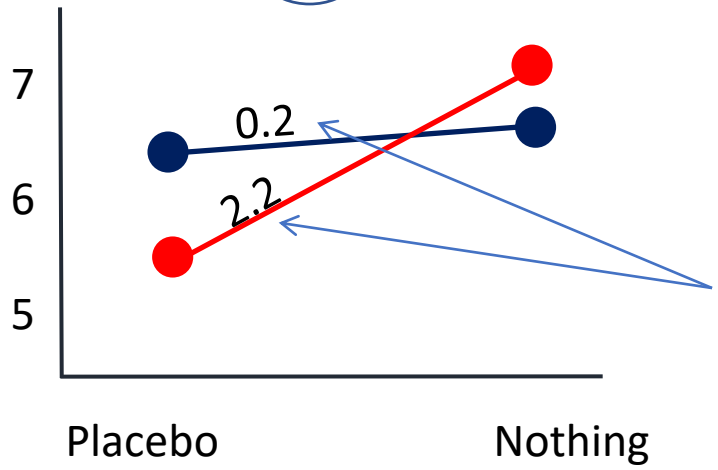
The slope of the blue line is the placebo effect among women

Plotting two-way factorial results

	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6

Difference of 2.2

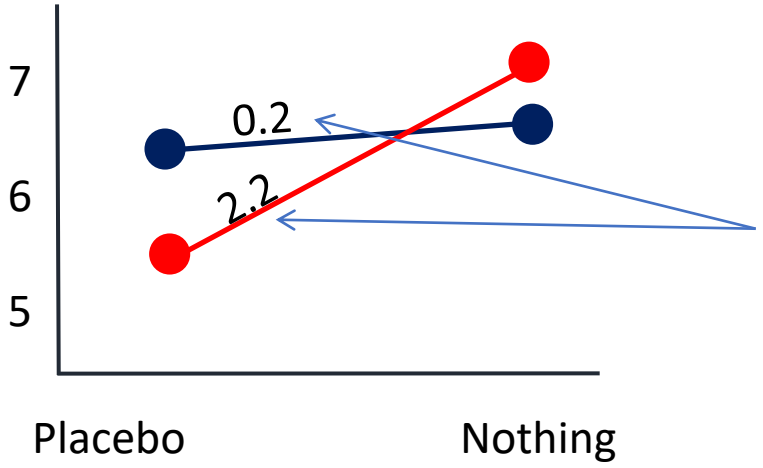
Difference of 0.2



Different slopes mean the effect is different for men than women – an INTERACTION!

Plotting two-way factorial results

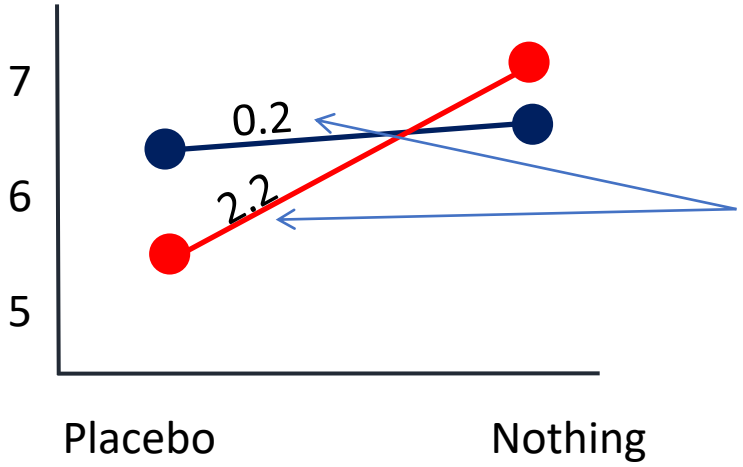
	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6



<u>Interaction definition</u>: the effects of one factor (placebo here) *depend* on the level of the other factor (gender here).

Plotting two-way factorial results

	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6



The degree to which lines are <u>not parallel</u> tells you how strong the interaction is.

Plotting two-way factorial results

	Men	Women	
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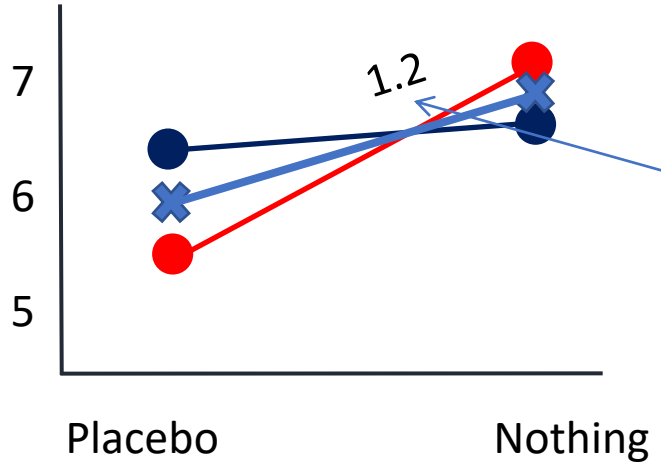
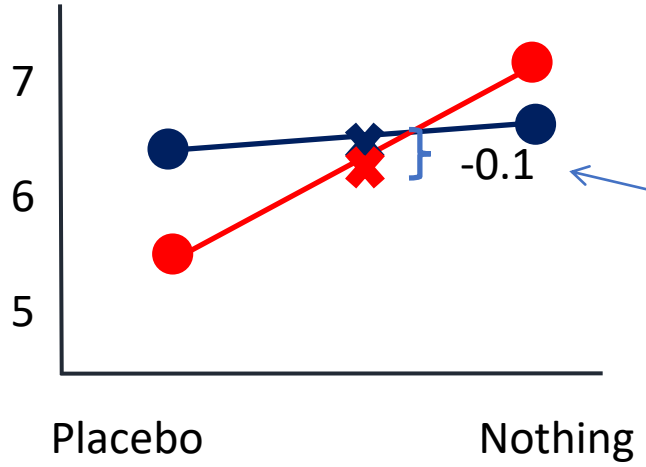


Main effect of placebo vs. nothing

Plotting two-way factorial results

	Men	Women
Placebo	5.3	6.4
Nothing	7.5	6.6
	6.4	6.5



Main effect of Men vs. Women

Practice at interpreting two-way factorial results:

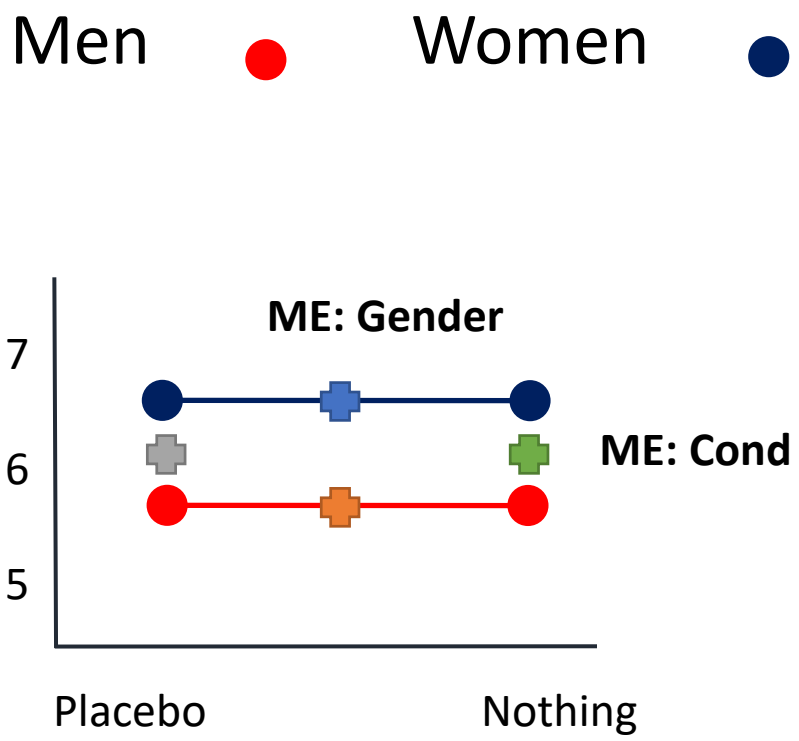
What is the main effect of gender? / What does that mean?

What is the main effect of cond? / What does that mean?

How do these terms interact? / What does that mean?

	P-F	P-M	N-F	N-M
λ1	1	-1	1	-1
λ2	1	1	-1	-1
λ3	1	-1	-1	1

$$Model\ A: y_i = \beta_0 + \beta_1(X_1) + \beta_2(X_2) + \beta_3(X_3) + \epsilon_i$$



Interaction: Check the slopes. Are the differences different?

Practice at interpreting two-way factorial results:

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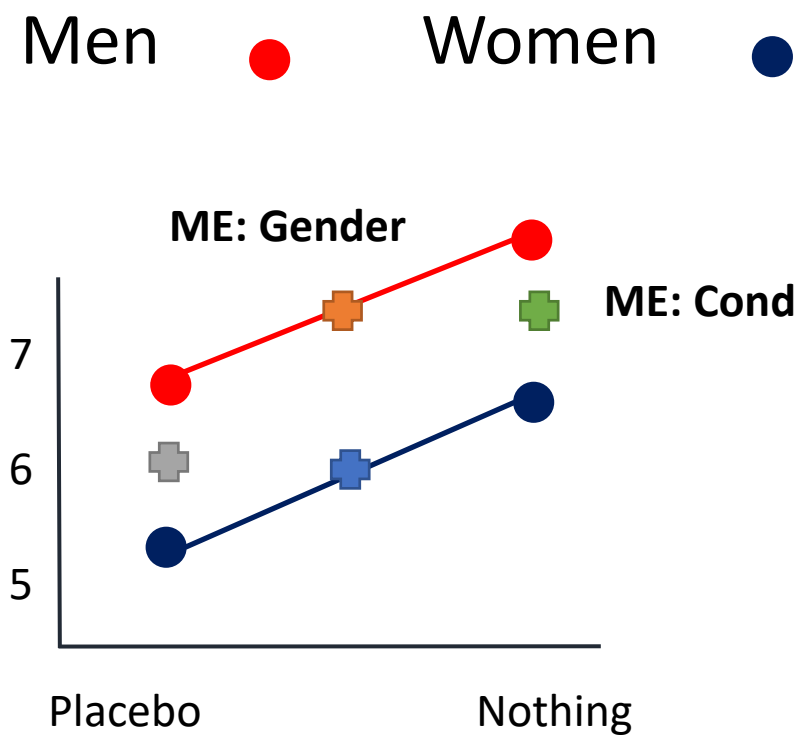

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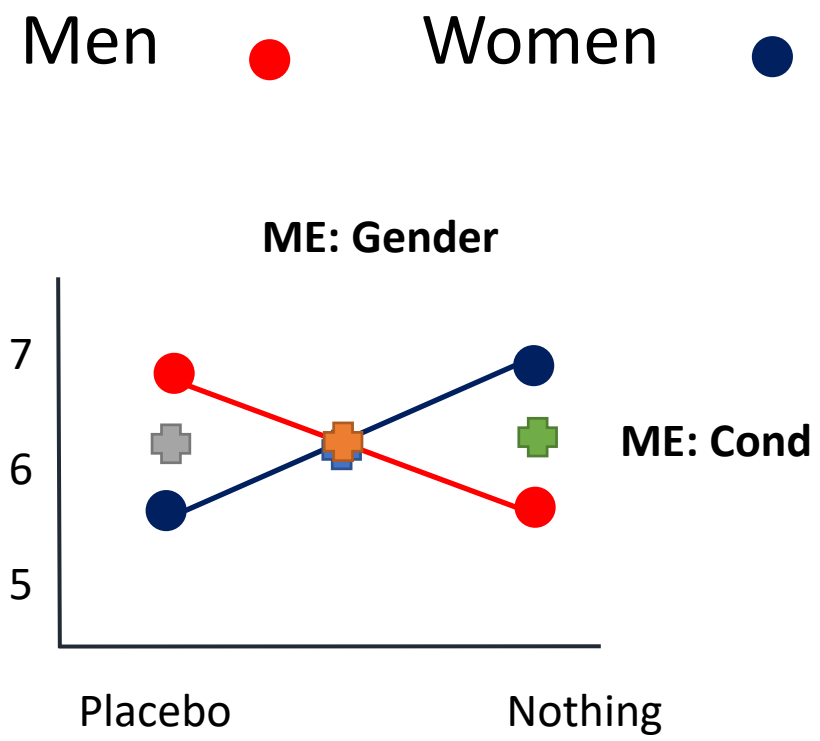

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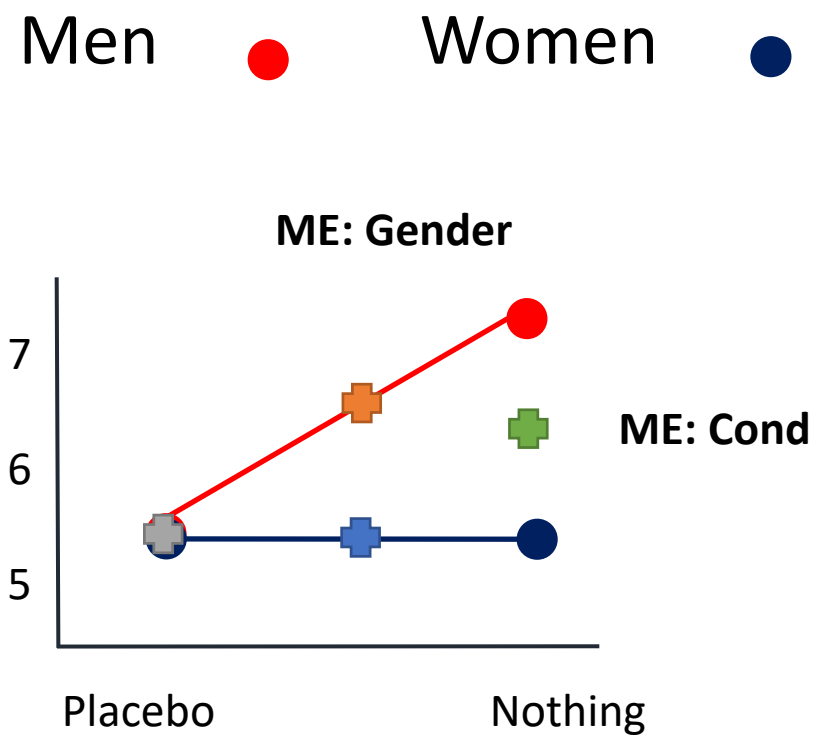

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